MA1522 Linear Algebra for Computing

Lesson 20: Range and Kernel of Linear Transformations

Announcements

- Today is the last lesson of new material.
- The remaining lessons are revision:
  - Fri 6 Nov: Chapters 1 & 2,
  - Tue 10 Nov: Chapters 3 & 4,
  - Fri 13 Nov: Chapters 5–6.
- **Final Exam**: **Tuesday 24 November 2026, 5:00 pm**. Closed book; one helpsheet is permitted. See the **Instruction of Final Exam** file on Canvas.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson20` (e.g. `A1234567Z_lesson20.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 20**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Section 7.1 Recall

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation.

- The **standard matrix** of T is $A = \begin{pmatrix} T(\mathbf{e}_1) & \cdots & T(\mathbf{e}_n) \end{pmatrix}$, and $T(\mathbf{u}) = A\mathbf{u}$ for all $\mathbf{u}$ in $\mathbb{R}^n$.
- If $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a basis for $\mathbb{R}^n$, then

$$A = [T]_S \, P_{E \to S} = \begin{pmatrix} T(\mathbf{u}_1) & \cdots & T(\mathbf{u}_n) \end{pmatrix} \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}^{-1}.$$

- If $\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_k\mathbf{v}_k$, then $T(\mathbf{v}) = c_1 T(\mathbf{v}_1) + \cdots + c_k T(\mathbf{v}_k)$. So the values of T on $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ determine T on their span.

Remark

Suppose that $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is a basis of $\mathbb{R}^n$, and $\{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is an arbitrary set of vectors of $\mathbb{R}^m$. Then there exists a unique linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$T(\mathbf{v}_i) = \mathbf{u}_i \qquad \text{for } i = 1, \ldots, n.$$

Existence. Let $V = \begin{pmatrix} \mathbf{v}_1 & \cdots & \mathbf{v}_n \end{pmatrix}$ and $U = \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_n \end{pmatrix}$. The columns of the square matrix V form a basis, so V is invertible. Define $T(\mathbf{x}) = UV^{-1}\mathbf{x}$; it is linear, being multiplication by the $m \times n$ matrix UV^{-1}. Since $V\mathbf{e}_i = \mathbf{v}_i$, we have $V^{-1}\mathbf{v}_i = \mathbf{e}_i$, so $T(\mathbf{v}_i) = UV^{-1}\mathbf{v}_i = U\mathbf{e}_i = \mathbf{u}_i$.

Uniqueness. Suppose T' is also linear with $T'(\mathbf{v}_i) = \mathbf{u}_i$. Since the $\mathbf{v}_i$ span $\mathbb{R}^n$, every $\mathbf{x} \in \mathbb{R}^n$ is $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$, and by linearity $T'(\mathbf{x}) = c_1\mathbf{u}_1 + \cdots + c_n\mathbf{u}_n = T(\mathbf{x})$. So $T' = T$. $\square$

Section 7.2 Range and Kernel

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A.

Definition

- The **range** of T is $R(T) = \{ \mathbf{v} \in \mathbb{R}^m \mid \mathbf{v} = T(\mathbf{u}) \text{ for some } \mathbf{u} \in \mathbb{R}^n \}$.
- The **rank** of T is $\text{rank}(T) = \dim(R(T))$.
- The **kernel** of T is $\ker(T) = \{ \mathbf{u} \in \mathbb{R}^n \mid T(\mathbf{u}) = \mathbf{0} \}$.
- The **nullity** of T is $\text{nullity}(T) = \dim(\ker(T))$.

Theorem

$R(T) = \text{Col}(A)$ is a subspace of $\mathbb{R}^m$, and $\ker(T) = \text{Null}(A)$ is a subspace of $\mathbb{R}^n$. Hence

$$\text{rank}(T) = \text{rank}(A), \qquad \text{nullity}(T) = \text{nullity}(A), \qquad \text{rank}(T) + \text{nullity}(T) = n.$$

Question 1

Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be the linear transformation

$$T\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 + x_3 + 3x_4 \\ -3x_1 - 6x_2 - 5x_3 - 11x_4 \\ -4x_1 - 8x_2 - 6x_3 - 14x_4 \end{pmatrix}.$$

1. Find a basis for $\ker(T)$, and write down $\text{nullity}(T)$.

2. Find a basis for $R(T)$, and write down $\text{rank}(T)$. Check that $\text{rank}(T) + \text{nullity}(T) = 4$.

Question 1 — solution: part 1

1. Reduce the standard matrix A: $R_2 + 3R_1$, $R_3 + 4R_1$; then $R_3 - R_2$, $-\frac{1}{2}R_2$, $R_1 - R_2$:

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ -3 & -6 & -5 & -11 \\ -4 & -8 & -6 & -14 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 & 3 \\ 0 & 0 & -2 & -2 \\ 0 & 0 & -2 & -2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

With $x_2 = s$, $x_4 = t$: $x_1 = -2s - 2t$, $x_3 = -t$, so

$$\ker(T) = \text{Null}(A) = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right\}, \qquad \text{nullity}(T) = 2.$$

Check: $A(-2, 0, -1, 1)^T = (-2 - 1 + 3,\ 6 + 5 - 11,\ 8 + 6 - 14)^T = \mathbf{0}$. ✓

Question 1 — solution: part 2

2. The pivot columns of the RREF are columns 1 and 3, so columns 1 and 3 of A form a basis for $R(T) = \text{Col}(A)$:

$$R(T) = \text{span}\left\{ \begin{pmatrix} 1 \\ -3 \\ -4 \end{pmatrix}, \begin{pmatrix} 1 \\ -5 \\ -6 \end{pmatrix} \right\}, \qquad \text{rank}(T) = 2.$$

$$\text{rank}(T) + \text{nullity}(T) = 2 + 2 = 4 = \text{number of columns of } A. \checkmark$$

$R(T)$ is a plane in $\mathbb{R}^3$, so T does not reach every vector of $\mathbb{R}^3$. $\square$

Challenge 1

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation, and let $[T]$ denote its standard matrix.

Show that $\text{Row}([T])$ is the orthogonal complement of $\ker(T)$ in $\mathbb{R}^n$.

Section 7.2 Injective and Surjective

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation with standard matrix A.

Definition

T is **injective**, or one-to-one, if $T(\mathbf{u}_1) = T(\mathbf{u}_2)$ implies $\mathbf{u}_1 = \mathbf{u}_2$.

T is **surjective**, or onto, if for every $\mathbf{v}$ in $\mathbb{R}^m$ there is a $\mathbf{u}$ in $\mathbb{R}^n$ with $T(\mathbf{u}) = \mathbf{v}$, that is, $R(T) = \mathbb{R}^m$.

Theorem

- T is injective $\iff$ $\ker(T) = \{\mathbf{0}\}$ $\iff$ $\text{rank}(A) = n$.
- T is surjective $\iff$ $R(T) = \mathbb{R}^m$ $\iff$ $\text{rank}(A) = m$.
- If $m = n$: T is injective $\iff$ T is surjective $\iff$ A is invertible.

Question 2

True or false? Justify your answer.

(a) There is an injective linear transformation $T : \mathbb{R}^5 \to \mathbb{R}^3$.

(b) If $T : \mathbb{R}^3 \to \mathbb{R}^5$ is linear and the reduced row-echelon form of its standard matrix has 2 nonzero rows, then T is injective.

(c) If $T : \mathbb{R}^3 \to \mathbb{R}^3$ is linear and $T\begin{pmatrix}3\\0\\1\end{pmatrix} = T\begin{pmatrix}1\\4\\-2\end{pmatrix}$, then T is not surjective.

Question 2 — solution

(a) False. By the rank-nullity theorem, $\text{nullity}(T) = 5 - \text{rank}(T) \geq 5 - 3 = 2$, so $\ker(T) \neq \{\mathbf{0}\}$ and T is not injective.

(b) False. The number of nonzero rows of the reduced row-echelon form is the rank, so $\text{rank}(T) = 2$ and $\text{nullity}(T) = 3 - 2 = 1$. Hence $\ker(T) \neq \{\mathbf{0}\}$.

(c) True. By linearity,

$$T\begin{pmatrix} 2 \\ -4 \\ 3 \end{pmatrix} = T\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix} - T\begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} = \mathbf{0},$$

so $(2, -4, 3)^T$ is a nonzero vector in $\ker(T)$ and T is not injective. Since the standard matrix is square, T is not surjective either. □

Question 3

For a real number k, let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 3y - 2z \\ 2x + 7y - z \\ 4x + 11y + kz \end{pmatrix}.$$

1. Find all values of k for which T is injective. For the same k, is T surjective?

2. For the value of k where T is not injective, find a basis for $\ker(T)$, and find an equation in v_1, v_2, v_3 such that $\mathbf{v} = (v_1, v_2, v_3)^T$ is in $R(T)$ if and only if the equation holds.

Question 3 — solution: part 1

1. T is injective if and only if its standard matrix A is invertible, if and only if $\det A \neq 0$. Expand along row 1:

$$\det \begin{pmatrix} 1 & 3 & -2 \\ 2 & 7 & -1 \\ 4 & 11 & k \end{pmatrix} = 1\,(7k+11) - 3\,(2k+4) + (-2)(22-28) = k + 11.$$

So T is injective if and only if $k \neq -11$.

For these k, T is also surjective: its standard matrix is square and invertible, so $\operatorname{rank}(T) = 3$ and $R(T) = \mathbb{R}^3$.

Question 3 — solution: part 2

2. Let $k = -11$. Reduce $(A \mid \mathbf{v})$: $R_2 - 2R_1$, $R_3 - 4R_1$; then $R_3 + R_2$:

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & v_1 \\ 2 & 7 & -1 & v_2 \\ 4 & 11 & -11 & v_3 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 3 & -2 & v_1 \\ 0 & 1 & 3 & v_2 - 2v_1 \\ 0 & -1 & -3 & v_3 - 4v_1 \end{array}\right) \to \left(\begin{array}{ccc|c} 1 & 3 & -2 & v_1 \\ 0 & 1 & 3 & v_2 - 2v_1 \\ 0 & 0 & 0 & -6v_1 + v_2 + v_3 \end{array}\right).$$

With $\mathbf{v} = \mathbf{0}$ and $R_1 - 3R_2$: $x - 11z = 0$, $y + 3z = 0$, so $\ker(T) = \operatorname{span}\left\{(11, -3, 1)^T\right\}$.

$\mathbf{v} \in R(T)$ if and only if the system is consistent, if and only if $-6v_1 + v_2 + v_3 = 0$. Check: column 1 of A gives $-6 + 2 + 4 = 0$. ✓ □

Challenge 2

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation, and let $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ be the standard basis for $\mathbb{R}^n$. Let $V_i = \text{span}\{\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_i\}$, so we have a filtration

$$\{\mathbf{0}\} \subset V_1 \subset V_2 \subset \cdots \subset V_n = \mathbb{R}^n.$$

Show the following.

1. The standard matrix $[T]$ is upper triangular if and only if T preserves the subspaces V_i; namely, for every i and every $\mathbf{v} \in V_i$, $T(\mathbf{v}) \in V_i$.

2. The standard matrix $[T]$ is orthogonal if and only if for all $\mathbf{v}_1, \mathbf{v}_2$ in $\mathbb{R}^n$,

$$T(\mathbf{v}_1) \cdot T(\mathbf{v}_2) = \mathbf{v}_1 \cdot \mathbf{v}_2.$$

Next lesson

Next lesson (**Fri 6 Nov**), we begin the revision with

1. Chapter 1: Linear Systems, and
2. Chapter 2: Matrix Algebra.

Please bring the questions you would like to go through.